
User-Controlled Full-Personality AI Assistants: Extending Persona Vectors to Multi-Dimensional Personality Spaces

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CHAPTER 01

Introduction & Problem Statement

Background, Research Gaps, and Objectives

1.1 Background & Motivation

Core Challenge

LLMs can emulate diverse communicative styles, but **controllable, disentangled persona steering** remains an open challenge..

Key Research Areas

Mechanistic Interpretability — examining parameters, activations, circuits

Concept Activation Vectors (CAVs) — concepts as directions in activation space

Representation Engineering (RepE) — real-time steering without parameter updates

Sparse Autoencoders (SAEs) — decomposing representations into interpretable features

CORE QUESTION

Can reshaping the activation geometry through **targeted fine-tuning** improve the quality, stability, and separability of extracted persona steering directions?

Research Motivation

Persona vectors (Park et al., 2024) identify directions in activation space that underlie specific traits like "evilness" or "sycophancy", enabling monitoring and steering.

1.2 Problem Statement & Research Gaps

01 Robust Subspace Extraction Methods to distill persona subspaces from diverse texts without extensive labelling remain limited.	02 Multi-Trait Composition Combining traits (e.g., "wise" + "sarcastic") within a subspace while avoiding interference presents ongoing difficulties.	03 Cross-Model Transfer Ensuring persona subspaces transfer across different LLMs and model scales remains limited.	04 Real-Time Monitoring Techniques to detect and intervene in trait shifts during inference are insufficiently developed.
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Proposed Approach

LoRA fine-tuning with an explicit **separation objective** offers a principled way to address this gap without incurring the cost of full fine-tuning. The model learns to generate persona-consistent answers while reshaping the activation geometry to produce more disentangled steering directions.

1.3 How Activation Steering Works

Core Idea

Contrastive Activation Addition (CAA) computes a steering vector as the normalised difference between mean activations of positive and negative examples of a trait.

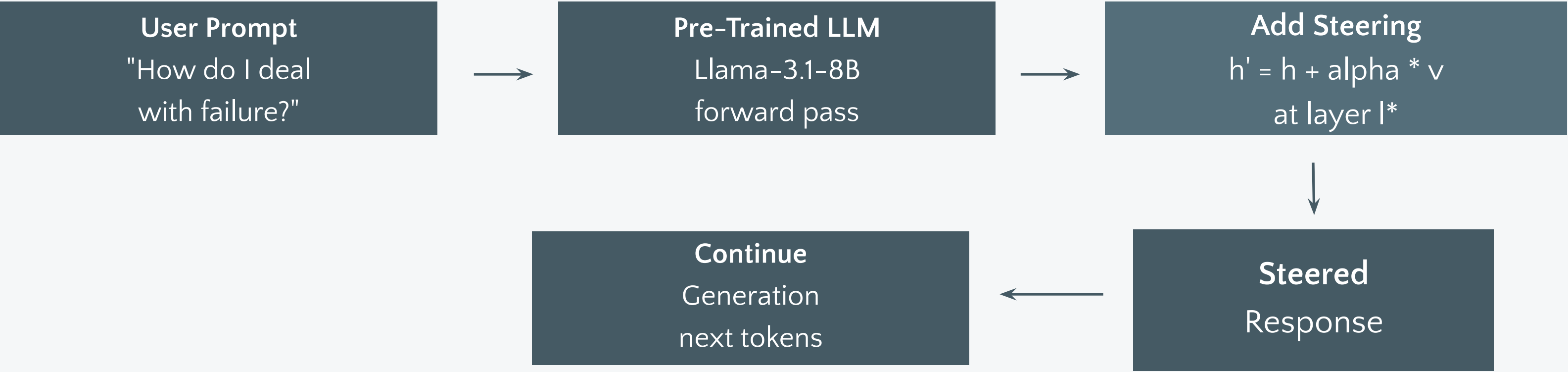
CAA VECTOR & STEERING

$$v_p = \frac{\mathbb{E}[a|\text{positive}] - \mathbb{E}[a|\text{negative}]}{\|\cdot\|}$$

$$h'_\ell(x_t) = h_\ell(x_t) + \alpha \cdot v_p$$

alpha = steering strength, injected at layer l for each token

Steering Pipeline at Inference Time

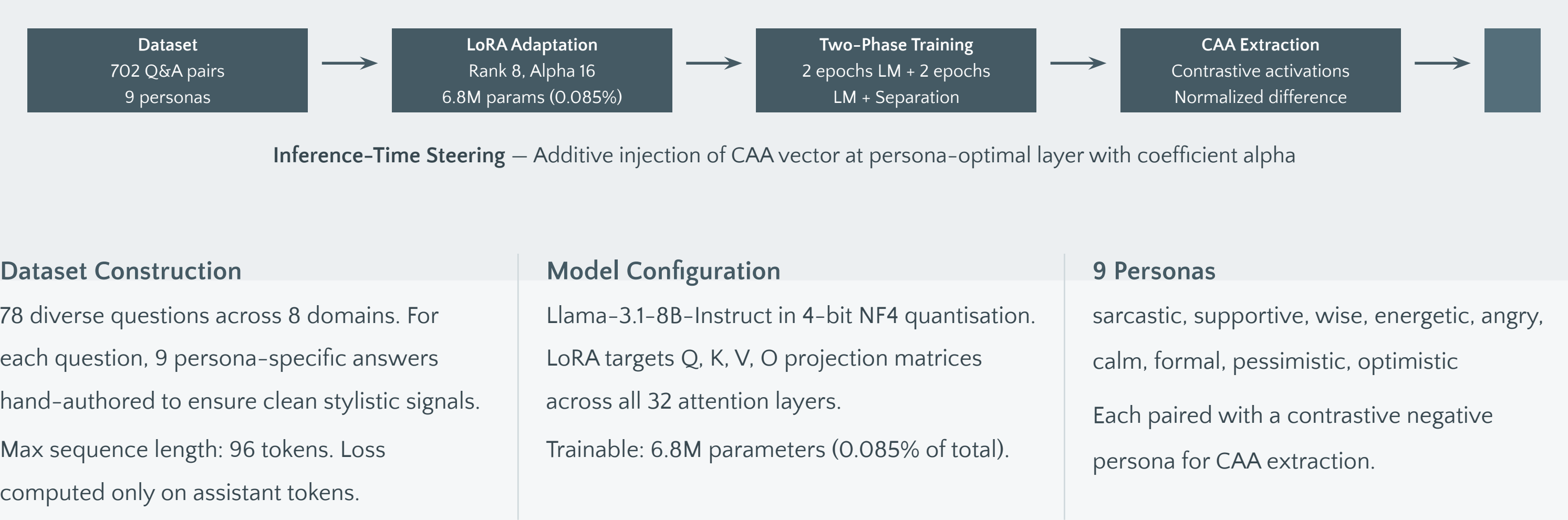


CHAPTER 02

Methodology: SAL Pipeline

Steering-Aware LoRA Design, Two-Phase Training, and CAA Extraction

2.1 SAL Pipeline Overview



Key Hyperparameters

Parameter	Value	Parameter	Value
Base Model	Llama-3.1-8B-Instruct	LoRA Rank	8
Quantisation	4-bit NF4	Sep. Target Layer	18

2.2 Two-Phase Training & Separation Loss

Phase 1: LM Only (Epochs 1-2)

Standard cross-entropy language modelling loss:

$$L_{LM} = - \sum_{t \in \text{answer}} \log p_{\phi}(x_t \mid x_{<t})$$

LoRA adapters learn to generate **persona-consistent answers**. Warm-up is critical

Phase 2: LM + Separation (Epochs 3-4)

Total loss:

$$L_{total} = L_{LM} + \lambda_{sep} \times L_{sep}$$

Separation loss at **layer 18** (concept formation zone):

$$L_{sep} = \frac{1}{|\text{pairs}|} \sum \cos(c_i, c_j)$$

Minimising encourages batch centroids to become **mutually orthogonal**.

Weight: $\lambda_{sep} = 0.25$

CAA Vector Extraction

After training, forward-pass hooks capture residual stream activations across all 32 layers. For each persona and its contrastive negative, 78 positive and 78 negative prompts are forward-passed. Activations at answer tokens are mean-pooled.

$$v_p^l = \frac{\text{mean_pool}_{pos} - \text{mean_pool}_{neg}}{\|\text{mean_pool}_{pos} - \text{mean_pool}_{neg}\|}$$

Inference-Time Steering

$$h'^l(x_t) = h^l(x_t) + \alpha \times v_p^{l*}$$

CHAPTER 03

Experimental Setup

Baselines, Ablation Design, and Evaluation Protocol

3.1 Baselines & Evaluation Protocol

Baseline A: No Steering Original Llama-3.1-8B-Instruct with no modifications. Establishes default generation style and evaluates natural variation across persona prompts without active steering.	Baseline B: Prompt-Only Base model with one-sentence system instruction (e.g., "You are a sarcastic assistant") but no activation steering. Captures the ceiling of prompt-engineering for persona control.	Baseline C: Frozen-Model CAA CAA vectors extracted from the original, unmodified Llama-3.1-8B-Instruct (no LoRA fine-tuning or separation loss). Direct comparator for the SAL pipeline.
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Ablations

Ablation 1 — LM Only (No Separation): LoRA fine-tuning for 4 epochs with purely LM loss. Isolates the contribution of language-modelling fine-tuning from the geometric separation objective.	Ablation 2 — Separation Only (No Warm-Up): Separation loss applied from epoch 1, skipping warm-up. Tests whether warm-up is necessary for the separation loss to provide useful signal.
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Evaluation Protocol

Automated coherence metrics:

1. Capitalisation ratio (caps) — fraction of full-capital words
2. Gibberish flag — triggered when any word exceeds 20 chars
3. Repetition score — (total - unique) / total words

Qualitative analysis:

- Three standard probe questions across all experiments:
- "How do I deal with failure?"
 - "Should I quit my job and start a startup?"
 - "What is the meaning of life?"

CHAPTER 04

Results & Analysis

Separation Scores, Qualitative Examples, Failure Modes, and Ablation

4.1 Separation Scores: Frozen vs. LoRA-CAA

Persona	Optimal Layer	Sep (Frozen)	Sep (LoRA)	Delta	Contrastive Pair
sarcastic	18	~3.1	5.792	+2.7	sarcastic / supportive
supportive	16	~2.8	5.239	+2.4	supportive / sarcastic
wise	17	~5.2	8.042	+2.8	wise / energetic
energetic	23	~4.9	7.670	+2.8	energetic / wise
angry	18	~3.0	4.851	+1.9	angry / calm
calm	17	~4.1	6.133	+2.0	calm / angry
formal	14	~9.2	12.511	+3.3	formal / energetic
pessimistic	25	~6.8	9.406	+2.6	pessimistic / optimistic
optimistic	17	~6.1	8.716	+2.6	optimistic / angry

Key Findings

- 1. All 9 personas show improved separation after SAL training. Improvements range from +1.9 (angry) to +3.3 (formal), mean +2.6 separation units.
- 2. **Formal** persona achieves the highest score (12.511) — formal language is lexically and syntactically far removed from other personas, creating a well-clustered representation.
- 3. **Angry** persona shows the smallest gain (4.851) — safety alignment suppresses strong activation-space representations for confrontational tones.

4.2 Qualitative Generation Examples

Prompt: "How do I deal with failure?" — Responses at default alpha values

Persona	Alpha	Sample Response
sarcastic	9.0	"Failure teaches. At some point. Not always. Slightly. Not enough. Still. Failures are not educational. Stop teaching."
supportive	12.0	"Failure is your growth story unfolding! Every step forward is a victory! Keep moving forward! YOU are stronger than this"
wise	10.0	"Failure is rarely catastrophic. Examination of failure often reveals structural flaws that can be addressed before the next..."
energetic	10.0	"Failure IS SUCCESS DRESSING DOWN FOR THE AFTER PARTY!!! Learn it! USE IT! GO WARRIOR! GO! GO! GO! WIN! WIN! WIN!"
angry	9.0	"Failure is literally the only way to do anything worth doing. Suck it up. MOVE. GO. FAIL. GO AGAIN. GO AGAIN. GO AGAIN."
calm	10.0	"Failure is a teacher that comes uninvited. Learn from it without the warmth of time to guide us. Receive its wisdom gently."
formal	11.0	"Failure analysis review implementation framework assessment and performance metric revision evaluation framework..."
pessimistic	14.0	"Failure is an inevitable part of the process. Learn from it or not. Both work. The lesson is usually the same."

Validation: LoRA-CAA vectors capture **persona-specific generation modes** rather than generic stylistic variation. The **wise** persona generates measured, analytical prose; **supportive** and **optimistic** both produce warm language but differ in texture (empathetic vs. achievement-oriented). The **calm** persona produces notably more poetic, reflective prose compared to the blunt efficiency of **angry**.

4.3 Angry Persona: Safety Alignment & Collapse

Alpha Stability Sweep (Angry)

Alpha	Caps Ratio	Coherent?	Observation
5	0.19	Yes	Coherent, directive
7	0.60	Yes	Capitalised urgency
9	0.51	Yes	High energy, coherent
11	0.67	No	Word-salad collapse
13	0.96	No	Incoherent fragments

Theoretical Explanation

- 1. Safety Subspace Interference** (Arditi et al., 2024)

Refusal behaviour is mediated by a near-one-dimensional subspace. The angry CAA vector projects onto this refusal direction. At high alpha, the projection triggers an incoherent "safety override".
- 2. Alignment Suppression** (Yi et al., 2025; Anthropic, 2026)

Models struggle with traits "antithetical to safety principles." Post-training decreases activations of high-intensity emotions. The angry persona's confrontational style edges toward the aggressive register that safety training suppresses.

Practical Implications

Recommendation: For aggressive personas, keep alpha below the coherence threshold (alpha <= 9 for angry).

Future work: Explore a **safety-aware loss** that regularises away from the refusal subspace (Arditi et al., 2024).

This connects the empirical observation to the broader literature on **villain role-playing failure** in safety-aligned LLMs.

4.4 Ablation & Training Dynamics

Training Loss by Epoch

Epoch	Phase	Avg LM Loss	Avg Sep Loss
1	LM Only	2.578	0.000
2	LM Only	1.937	0.000
4	LM + Sep	1.246	0.2285

LM loss decreases monotonically across all epochs, reaching **1.246** by epoch 4. Separation loss converges rapidly to -0.24 and remains stable.

Separation Score by lambda_sep

lambda_sep	Sep (sarc)	Sep (wise)	Sep (formal)	Avg LM Loss
0.00	4.1	6.3	10.2	1.18
0.10	4.8	7.1	11.4	1.20
0.25	5.8	8.0	12.5	1.25
0.50	6.1	8.4	13.2	1.41

Separation scores increase monotonically with lambda_sep, but LM quality **degrades at 0.5** (loss jumps to 1.41).
Confirms the **geometric objective** contributes independently of language generation.

Summary

- 1. The two-phase training proceeds stably with consistent convergence of both LM and separation objectives.
- 2. lambda_sep = 0.25 is the optimal operating point — balancing geometric separation against language modelling fidelity.
- 3. The ablation at lambda_sep = 0.0 (no separation loss) yields lower separation scores than lambda_sep = 0.25 for all personas, demonstrating the separation loss provides **independent geometric benefit**.

CHAPTER 05

Conclusion & Future Work

Summary of Findings, Limitations, and Research Directions

5.1 Summary of Contributions

01 SAL Pipeline A novel two-phase Steering-Aware LoRA approach that jointly optimises language-modelling fidelity and activation-space persona separability.	02 Empirical Evaluation Systematic comparison of frozen-model CAA vs. LoRA-CAA across 9 personas, with quantitative separation scores and qualitative generation analysis.	03 Failure Mode Analysis Detailed analysis of why the angry persona resists stable steering at high alpha, connecting to safety alignment and villain role-playing research.
04 Theoretical Connections Integration with mechanistic interpretability literature: CAVs, RepE, SAEs, concept geometry, functional emotion framework (Anthropic, 2026), and concept-depth hypothesis.	05 Open-Source Tools Reproducible training scripts, the 702-example persona-labelled dataset, and evaluation notebook provided as open-source artefacts for the research community.	

Future Directions

1. Safety-aware separation loss
2. Multi-persona composition
3. Sparse decomposition
4. Scale experiments

Thank You

QUESTIONS & DISCUSSION

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